

The Tail Is the Only Signal: Flush Reversion in Equity Indices and Crypto Perpetual Futures

Why pooled event studies overstate short horizon reversal, and what survives when they are corrected

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Working paper. Preliminary. Comments welcome. This version: August 2026.

JEL classification: G14, G12, G11, C12, C58

Keywords: short horizon reversal, event study, cross sectional dependence, liquidity provision, multiple testing, familywise error, stationary bootstrap, cryptocurrency perpetual futures, liquidation cascades, negative results

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Abstract

Buying an asset immediately after a large single day decline is one of the oldest ideas in short horizon trading, and it is easy to make it look profitable. This paper applies one event definition and one testing protocol to two asset classes, four US equity indices over seventy six years and, in results carried over from an earlier stage of the same research program, twenty cryptocurrency perpetual futures, and finds that almost all of the apparent effect is an artifact of how the events are counted. Large declines arrive on the same few calendar dates across instruments, so an event study that treats each instrument day as an independent observation is counting one market wide shock many times. In a panel of twenty eight single names, 1,361 declines at or below seven percent are only 791 distinct dates, and the pooled t statistic of 6.76 falls to 1.68 when each date is counted once. Three further corrections matter and are usually skipped: the depth thresholds that define such events are nested rather than independent, so a strong tail leaks into every shallower cut; testing several depths requires a familywise correction; and a stationary block bootstrap with a block length that is a large fraction of the sample is badly anti conservative,

which in our own first pass manufactured a p value two orders of magnitude too small. After all four corrections, exactly one result survives in equities. Index declines at or below seven percent, 50 events on 25 dates clustering into 10 episodes since 1950, are followed by a mean next session return of 3.08 percent, with a familywise p value of 0.006 and a bootstrap interval from 1.34 to 4.84 percent. It is not crash window drift: the next session returns 3.08 percent while each of the following nine averages 0.01 percent, below the unconditional daily mean. It is implementable, since 83 percent of the move occurs after the opening print. Everything shallower is indistinguishable from zero, and the corresponding single name effect is absent. In crypto perpetuals the same mechanism fails completely, with an edge net of market beta that is negative in every regime and every year of a six year sample. The economics are the final constraint: an account trading the surviving tail signal deploys 0.05 percent of its capital time and beats Treasury bills by 0.44 points a year, which is a real effect that cannot carry a portfolio.

1. Introduction

A large down day is an invitation. Prices have moved a long way in a few hours, someone was probably forced to sell, and whoever is willing to take the other side is being paid to supply liquidity when it is scarcest. The idea has a long history in the reversal literature and a simple trading expression: buy the close of a bad day and sell the next one. This paper asks how much of that idea is real, using one event definition and one testing protocol applied to two asset classes that could hardly be more different in their microstructure.

The answer is mostly a lesson about counting. Large declines are not scattered independently through time; they arrive together, because the thing that causes an index to fall five percent is usually the same thing that causes every other index and most individual stocks to fall on the same afternoon. An event study that pools instrument days therefore inflates its sample enormously without adding much information, and the t statistic it reports is a statement about how many instruments were sampled rather than about how many distinct shocks were observed. In the single name panel used here the effect is stark. At the deepest threshold there are 1,361 events, and they fall on 791 dates. Pooling them gives a t of 6.76, which would be decisive in any seminar. Counting each date once gives 1.68, an unadjusted p of 0.045 that does not survive a correction for the three thresholds tested (Figure 1).

Three further corrections are needed and are easy to omit. The depth thresholds normally used to define these events are nested: every decline of seven percent is also a decline of five percent and of three percent, so a genuine tail effect appears, diluted, in every shallower cut and can be mistaken for three independent confirmations. Testing several depths is a multiple comparison and needs a familywise correction. And the stationary block bootstrap, which is the natural tool for dependent data, becomes badly anti conservative when the mean block length is a large fraction of a small sample. Our own first pass fell into exactly that trap and reported a p value roughly two orders of magnitude too small. We describe the error rather than quietly fixing it, because the mechanism is general and the corrected protocol is the contribution.

After all four corrections, one equity result survives, and it is a narrow one. Index declines at or below seven percent, of which there have been 50 across four indices on 25 distinct dates since 1950, are followed by a mean next session return of 3.08 percent. That result clears a familywise correction, survives the removal of 2008 and 2020, and passes a placebo test that the shallower thresholds fail badly. Everything above the tail is indistinguishable from zero, and in crypto perpetual futures the same mechanism produces nothing at any depth, in any regime, in any year. The paper closes with the constraint that matters most in practice and is almost never reported: the surviving signal fires on twenty five days in seventy six years, so an account that trades it holds Treasury bills for 99.95 percent of its dollar days and beats them by less than half a point a year.

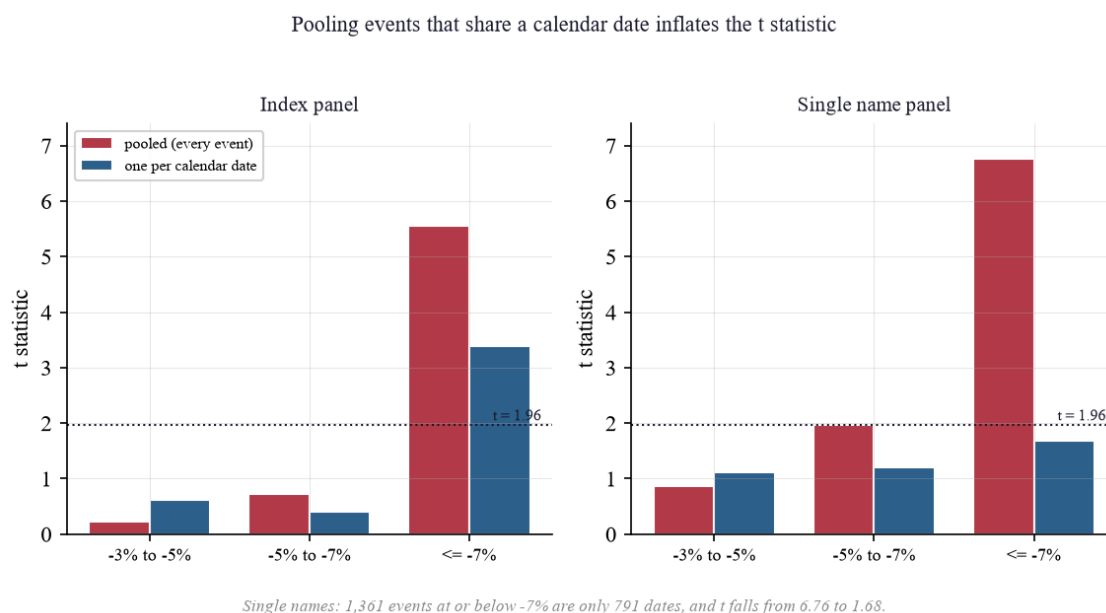


Figure 1. The counting problem. Pooling every instrument day inflates the t statistic because large declines share calendar dates. In the single name panel the pooled t of 6.76 at the deepest threshold falls to 1.68 once each date is counted once, moving the result from decisive to insignificant. The index panel is affected in the same direction but less severely, because it contains only four instruments.

2. Background

Short horizon reversal is a well documented feature of equity returns. Lehmann (1990) and Jegadeesh (1990) established that weekly and monthly losers tend to outperform in the period immediately following, and Lo and MacKinlay (1990) showed that a large part of measured contrarian profit comes from cross sectional structure rather than from any single security overreacting. The mechanism most consistent with the evidence is compensation for liquidity provision rather than correction of a behavioral error. Grossman and Miller (1988) modeled the price concession required to induce intermediaries to absorb an order imbalance, Campbell, Grossman and Wang (1993) tied return reversal to volume in a way that identifies it with non informational selling pressure, and Nagel (2012) showed directly that the returns to short horizon reversal strategies rise sharply when liquidity is scarce, behaving like the payoff to market making. Avramov, Chordia and Goyal (2006) and Da, Liu and Schaumburg (2014) refine where in the cross section the effect lives, and Hendershott and Menkveld (2014) measure the price pressure and its decay directly. Hameed, Kang and Viswanathan (2010) add the asymmetry that matters most for the present design: liquidity withdraws specifically after negative market returns rather than after large moves of either sign, and the withdrawal intensifies when funding is tight. Our tail result is consistent with this reading, since what survives is concentrated exactly where liquidity provision should be most constrained and most richly paid.

The methodological literature matters here as much as the economic one, and the central problem in this paper is a known one with a known name. Kolari and Pynnonen (2010) show that when event dates cluster, even low cross correlation among abnormal returns produces serious over rejection of the null, and they supply an adjusted statistic for it; Mitchell and Stafford (2000) had already demonstrated that pooled inference assuming independence across event firms is invalid, and that apparent abnormal performance can largely evaporate once positive cross correlation is admitted, which is the standard argument for moving to calendar time portfolios. Flush days cluster almost perfectly by construction, since a market wide decline strikes every instrument in

the same session, so this literature describes our setting exactly. Counting one observation per calendar date is the bluntest member of that family of remedies and the most transparent, which is why we use it. The multiple testing literature supplies the second correction. White (2000) and Romano and Wolf (2005) formalize the reality check and the stepwise familywise approach for exactly the situation where a researcher scans several specifications and reports the best, and Harvey, Liu and Zhu (2016) argue that conventional thresholds are far too permissive once the breadth of the search is admitted. Bailey and Lopez de Prado (2014) make the same argument in terms of a deflated Sharpe ratio. The third correction concerns the bootstrap itself: the stationary bootstrap of Politis and Romano (1994) is the right tool for dependent series, but its block length is a tuning parameter, and choosing one that is a large fraction of a short sample destroys the test's calibration in the anti conservative direction.

For the crypto side the relevant background is structural rather than statistical. Perpetual futures are leveraged instruments with automatic liquidation, so a decline mechanically produces forced selling by the exchange itself, and the resulting cascade is a different object from an equity market maker absorbing inventory. Brunnermeier and Pedersen (2009) describe the general mechanism by which funding constraints and market liquidity reinforce each other, which is the closest theoretical description of what a liquidation cascade does. Makarov and Schoar (2020) document the frictions and segmentation that make these markets behave unlike mature equity markets. The empirical question this paper puts to that literature is simple: if reversal after forced selling is compensation for supplying liquidity, does it appear in a market where the forced selling is larger, faster, and fully automated? It does not.

3. Data and Method

The equity data are daily open and close prices from a public vendor API for four US indices, the S&P 500 from 1950, the Nasdaq Composite from 1971, the Russell 2000 from 1987 and the Dow Jones Industrial Average from 1992, together with a panel of twenty eight large capitalization single names. The union of index trading days gives a fixed calendar of 19,276 sessions from 1950-01-04 to 2026-08-17, a span of 76.6 years. The risk free rate is the thirteen week Treasury bill series, used both to accrue idle cash in the account simulation and to compute excess returns. An earlier version of this analysis began in 1990 and therefore excluded 1987-10-

19, the largest single day decline in the record; the start date was moved back once that exclusion was recognized as a choice rather than a constraint.

The event is deliberately crude, because the point is to test the idea in its simplest form. A flush is a daily close to close return at or below a threshold. The trade is to buy that close and sell the next one, with a twenty basis point round trip cost. Three thresholds are examined, at three, five and seven percent, and they are treated as disjoint bands rather than nested cuts: the first band contains declines between three and five percent, the second between five and seven, and the third everything at or below seven. This is the single most consequential design choice in the paper. Under the conventional nested definition the tail contaminates every shallower threshold, and three correlated views of one result present themselves as three confirmations.

Inference follows four rules. Each calendar date contributes one observation, the mean across whatever instruments flushed that day, which is the direct remedy for cross sectional dependence. Confidence intervals and p values come from an independent and identically distributed bootstrap on those date level observations, with ten thousand resamples; block bootstrap variants are refused outright whenever the sample contains fewer than twenty blocks, for reasons documented in Section 4. Because three bands are tested, a familywise max statistic correction is applied across them by resampling the recentred date series and comparing each observed statistic to the distribution of the maximum. Finally, dates are clustered into episodes when they fall within sixty days of one another, and an episode level statistic is reported alongside the date level one, because twenty five dates drawn from ten crises are not twenty five independent draws. The episode statistic is reported for the index panel only; in the much denser single name panel the clustering rule degenerates and the statistic is not meaningful.

The account simulation walks the entire fixed calendar rather than only the event dates. On a flush date capital is spread equally across that date's signals, capped at twenty percent per position and ninety five percent in aggregate; on every other day the account earns the Treasury bill rate. Walking the full calendar matters for three separate reasons. It prevents the annualization window from being chosen by the signal itself, it stops idle capital from being silently free, and it allows exposure to be reported in capital time, the fraction of dollar days actually at risk, rather than in the far more flattering unit of days on which a trade happened to occur.

4. The Pooled Statistic Is an Artifact

The first result is that most of the apparent significance in this literature's simplest design comes from counting. Table 1 gives both statistics for both panels. In the index panel the pooled t at the deepest band is 5.55 and the date level t is 3.38, a substantial but survivable haircut, which is what one expects when only four instruments can share a date. In the single name panel the same comparison runs from 6.76 to 1.68. Twenty eight instruments crashing together on 791 days produce 1,361 events, and the pooled statistic reads the multiplicity of instruments as though it were the multiplicity of evidence.

Panel and band	Events	Dates	Pooled t	Date level t	Bootstrap p (uncorrected)
Indices, -3% to -5%	535	322	0.22	0.61	0.271
Indices, -5% to -7%	88	55	0.73	0.40	0.327
Indices, at or below -7%	50	25	5.55	3.38	0.0005
Singles, -3% to -5%	11,556	5,231	0.87	1.11	0.138
Singles, -5% to -7%	2,545	1,580	1.97	1.21	0.117
Singles, at or below -7%	1,361	791	6.76	1.68	0.045

Table 1. Pooled and date level statistics for both panels, on disjoint bands. The single name panel is where the counting problem bites hardest: a pooled t of 6.76 becomes 1.68. Note also that the single name panel has roughly nineteen times as many distinct dates as the index panel, and thirty two times as many in the tail band, so its failure to produce a significant effect is a failure in the better powered sample, not a small sample result. The final column is uncorrected for the three bands tested; the familywise values appear in Table 2, where the index tail band's 0.0005 becomes 0.0058 and no other band survives.

It is worth being precise about what this does and does not show. It does not show that the single name effect is exactly zero; the point estimates are positive at every band. It shows that a panel with roughly nineteen times the index panel's number of distinct dates, and therefore far more power, cannot reject the null at any depth once its events are counted honestly and the three bands are charged a multiplicity correction, its strongest band reaching only an unadjusted p of 0.045. That is a failed replication in the better powered sample, which is a considerably stronger statement than a null in a small one. Two known biases run in opposite directions here and

neither rescues the result: these twenty eight names are survivors, which if anything inflates measured post decline reversion, while the use of price rather than total return series slightly understates it.

The second counting problem is nesting. If the thresholds are defined as cumulative cuts, then every seven percent decline also appears in the five percent sample and the three percent sample. A tail effect of a few percent spread across a few hundred shallow events is easily large enough to push a diluted average across a significance threshold, at which point a researcher sees three thresholds agreeing and reads it as robustness. Under disjoint bands the picture inverts completely: the two shallower bands have date level statistics of 0.61 and 0.40, which is nothing at all.

The third problem concerns the bootstrap, and it is worth reporting in the first person because we made the error. The natural tool for possibly dependent event series is the stationary bootstrap, and our first pass used it with a mean block length of ten. That pass began in 1990 and used nested cuts, so its deepest band held twenty two date level observations rather than the twenty five of Table 2, and the mean block was forty five percent of that sample; circular resampling then degenerates toward rotations of the same series, the bootstrap standard error collapses, and the statistic inflates. The size of that test can be measured directly. Against genuine independent nulls at the sample sizes this paper actually uses, a mean block of ten rejects a true null 13.3 percent of the time at twenty two observations and 10.6 percent at fifty eight, against a nominal five percent; the independent bootstrap that replaced it rejects at 6.5 and 5.6 percent, which is calibration. A p value from the first configuration is therefore not interpretable at face value, whatever it reads. The corrected protocol uses an independent bootstrap on date level observations, refuses any block configuration with fewer than twenty blocks, and never prints a p value of zero, since ten thousand resamples cannot resolve below one in ten thousand. Two independent diagnostics confirm that no block correction was needed in the first place. The first order autocorrelation of the date level series is small at every band and negative at the two deeper ones, at 0.062, minus 0.137 and minus 0.112 going from the shallowest band to the tail, so there is little positive dependence for a block to capture. And blocking in event index space is not blocking in calendar time: the median gap between consecutive events is 10, 24 and 14 days by band while the largest gaps run to more than ten years, so adjacent entries in the series are frequently separated by an entire market cycle.

5. Equity Indices: Only the Tail

With the protocol fixed, the equity index result is easy to state. Two of the three bands are empty and one is large. Table 2 and Figure 2 give the full picture. Declines between three and five percent are followed by a mean next session return of 0.089 percent, against a bootstrap interval that comfortably contains zero and a familywise p value of 0.63. Declines between five and seven percent give 0.243 percent with a familywise p value of 0.73. Declines at or below seven percent give 3.075 percent, with a bootstrap interval from 1.335 to 4.835 percent and a familywise p value of 0.0058. Seventy two percent of those dates were followed by a positive session.

Band	Events	Dates	Episodes	Mean next session	Date t	Episode t	Familywise p
-3% to -5%	535	322	65	+0.089%	0.61	0.54	0.631
-5% to -7%	88	55	22	+0.243%	0.40	-0.04	0.733
At or below -7%	50	25	10	+3.075%	3.38	3.89	0.0058

Table 2. The index panel on disjoint bands, 1950 to 2026. Only the extreme tail survives, and it survives both the date level correction and a familywise correction across the three bands. The episode column counts clusters of dates falling within sixty days of one another: the surviving result rests on ten episodes in seventy six years, roughly one every seven and a half years.

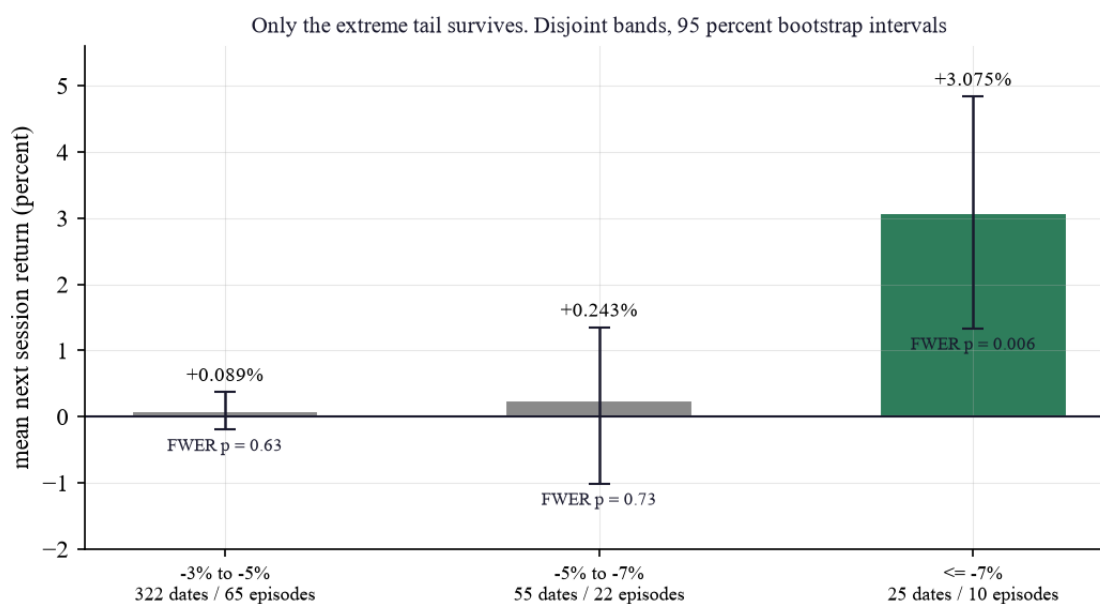


Figure 2. Effect size by disjoint band with ninety five percent bootstrap intervals on date level observations. The two shallower bands are indistinguishable from zero and do not survive the familywise correction. The tail

band is large, and its interval excludes zero by a wide margin, but it rests on twenty five dates drawn from ten episodes.

Two robustness checks decide whether this is worth reporting. The first is the more important, because the obvious objection to any post crash result is that markets simply drift upward after crashes and any window would show a gain. To test that, hold the event fixed and move the outcome: measure the single day return two days after the flush, then three, and so on out to ten, and compare each with the return on the next session. The result is decisive and is shown in Figure 3. In the tail band the next session returns 3.075 percent while the following nine average 0.013 percent each, which is below the unconditional daily mean of about 0.03 percent, a ratio of 234 to one. Whatever is happening is concentrated in a single session and is not general post crash drift. In the shallowest band the same test returns 0.8, meaning the next session is slightly worse than the placebo days, which is exactly what one should see when there is no effect.

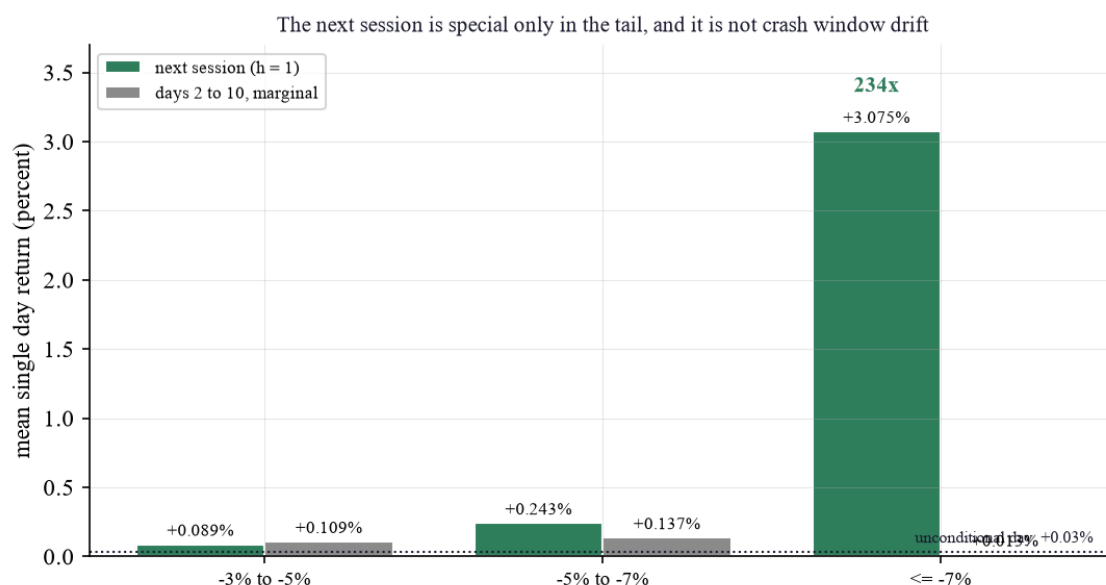


Figure 3. The placebo horizon test. Each bar compares the next session return with the average marginal single day return over the following nine sessions. In the tail the next session is 234 times the placebo and the placebo days sit below the unconditional daily mean, so the effect is not crash window drift. In the shallowest band the next session underperforms the placebo, consistent with no effect at all.

The second check removes the two years that dominate any modern crash sample. Excluding 2008 and 2020 leaves 20 events on 13 dates in 8 episodes, and the tail result holds: a mean of 3.009 percent, a date level t of 2.15, an episode level t of 3.47 and a bootstrap p value of 0.0095.

This is a genuine robustness result and it should be reported as such, but its limits deserve equal emphasis. A wider exclusion covering 1997, 1998, the 2000 to 2002 period, 2008, 2009, 2011, 2020 and 2022 leaves only three dates, on which the mean is 1.064 percent with a bootstrap p of 0.402. That is not a test, and we report it as a bound. A related caution is that an earlier analysis in this program found the per event edge to be flat across ex ante volatility regimes, so the correct statement is not that the effect is conditional on crisis but that the statistical significance of the pooled result depends on the crisis years, which is a different and weaker claim.

Finally, an effect that exists only between the closing print and the opening print of the next day would be a curiosity rather than a finding, since no one can trade it. Decomposing the tail move into the overnight gap and the regular session settles this, and the answer is favorable: the gap contributes 0.510 percent and the session contributes 2.567 percent, so 83 percent of the move is available to someone who waits for the market to open and buys there (Figure 6). The account simulation in Section 9 confirms the point end to end. Buying the next open instead of the flush close costs 23 percent of the per active day return, 1.349 percent falling to 1.032, but almost nothing in compounded terms, 4.26 percent a year falling to 4.16.

6. Single Names: A Failed Replication in the Larger Sample

If reversal after a large decline is compensation for supplying liquidity, it ought to be visible in individual stocks, where the reversal literature originally found it and where idiosyncratic selling pressure is least diluted. It is not visible here, and the sample is large enough that the absence carries weight. Across twenty eight names the three bands give date level statistics of 1.11, 1.21 and 1.68, with bootstrap p values of 0.138, 0.117 and 0.045. The deepest band is the only one approaching conventional significance, and at 0.045 across three tested bands it would not be expected to survive the familywise correction that reduced the index panel; that correction was computed for the index panel and is quoted here as an expectation, not as a measured result for this panel.

The interpretation requires care in two directions. First, this panel has 5,231, 1,580 and 791 distinct dates in the three bands, against the index panel's 322, 55 and 25. Power is not the limitation. Second, the panel is survivorship biased by construction, since it contains names that still trade today, and the direction of that bias is toward finding reversion and not away from it, because the firms that actually failed are precisely those whose large declines were never

followed by a recovery, and they are absent from the sample. We were not able to test this directly, since delisted securities are not retrievable from the data source used here, so the argument is a structural one about which observations are missing and not a measurement. It does mean the reported null is more likely conservative than generous. Offsetting it slightly, these are price series and not total return series, which understates realized reversion by the dividend accrued over one day, an amount too small to change any conclusion.

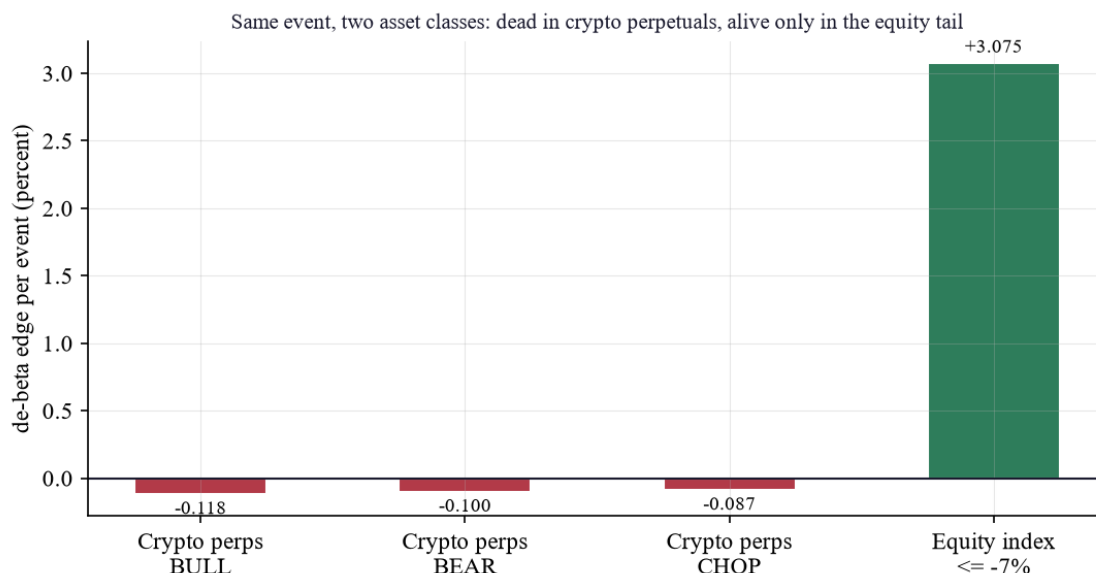
A reader inclined to defend the single name effect might argue that a five percent decline is a much milder event for a volatile individual stock than for an index, so absolute thresholds compare unlike things. That objection was tested during verification by replacing absolute cuts with standardized ones, estimated from the prior two hundred and fifty days only, so that no future information enters. Standardizing made the single name result weaker. At three trailing standard deviations the two panels are both flat, with date level statistics of 0.42 for the indices and 0.78 for the single names. At four the single name panel turns negative, minus 0.80 against 0.69. At six the two diverge completely: the index panel returns 2.784 percent on the next session with a date level statistic of 4.29, while the single name panel returns minus 0.618 percent with a statistic of minus 1.78 and wins on only 37.5 percent of its dates. Measuring severity in each instrument's own units therefore widens the asymmetry instead of explaining it away.

7. Crypto Perpetual Futures: Nothing at Any Depth

The crypto side of this comparison was measured earlier in the same research program, and the numbers here are cited from that dated record rather than recomputed for this paper; the Data and Code section states precisely which artifacts survive and which do not. The setting is twenty US dollar margined perpetual contracts on a major venue. A flush is defined analogously, outcomes are resolved on one minute prices against fixed brackets, and the sample runs from 2021 to 2026 and contains 87,870 events, with two caveats that matter. A native liquidation feed exists only from 2026-06-03, so the honest native span is about 63 days containing roughly 16 flush clusters and every earlier event is identified from price and volume. And that count of 87,870 is a pooled instrument day count, so the counting correction this paper applies to the equity panels has not been applied to the crypto one; since the crypto result is uniformly negative, that correction would only reduce an already absent effect.

The result is not a weak positive but a consistent negative. Measured as an edge net of market beta, flush reversion is negative in every market regime, at minus 0.118 in bull conditions, minus 0.100 in bear conditions and minus 0.087 in range bound conditions. Year by year across the six year sample, the estimated probability that the de beta edge is positive rounds to zero in every year, which given the resolution of that resampling exercise should be read as below one percent and not as an exact zero. The realized win rate of 47.9 percent sits below the geometry's breakeven. A machine learning meta labeler trained on the full state snapshot and evaluated with purged walk forward validation reaches an out of sample area under the curve between 0.493 and 0.506, which is a coin flip, and its confidence frontier is flat: the trades it is most confident about win 51.4 percent of the time against an average of 54.4 percent, and no confidence bucket reaches sixty percent.

Three further checks close off the usual escapes. Inverting the rule and trading continuation instead of reversion also loses, with directional win rates of 48.6 and 48.4 percent against a breakeven near 49.8, which is the signature of a coin flip paying fees in both directions rather than of a signal with the sign reversed. Changing the bracket geometry does not help, since net return per trade is identical at minus 0.09 percent across three different take profit and stop loss configurations, geometry sliding the win rate and the breakeven together. And the one genuinely large improvement found anywhere in that work was an execution effect rather than a prediction effect: a price improved limit entry turned out to be adverse selection in a falling market, filling only when the decline continued, and replacing it with a market order moved the baseline from minus 21.7 percent to plus 17.6 percent at an unchanged win rate. That is a finding about order types, not about reversion.



Crypto: 87,870 proxy events 2021 to 2026, de-beta edge negative in every regime and every year (cited, not re-run). Scales differ; the sign is the point.

Figure 4. The same event in two asset classes. In crypto perpetuals the de beta edge per event is negative in every regime and in every year of a six year sample. In equity indices the effect is absent at shallow depths and large in the tail. The vertical scales measure different quantities and are not directly comparable; the sign and the consistency are what the comparison rests on.

8. Why the Two Asset Classes Differ

The natural reading of the equity tail result is the liquidity provision story: a seven percent index day is an event in which the supply of intermediation is overwhelmed, and the next session pays whoever was willing to stand in front of it. On that reading crypto perpetuals should pay more, not less, since the forced selling is larger relative to available depth and arrives faster. They do not, and the difference deserves a careful statement.

The most likely explanation is that the two events are not the same event. In an equity market the seller in a crash is a human or an institution making a decision under stress, and the intermediary who absorbs the inventory earns a concession for holding it overnight. In a perpetual futures market a large decline mechanically triggers automatic liquidation of leveraged positions, and the liquidation engine is itself a seller that does not care about price. The magnitudes are not marginal: Cheng, Deng, Wang and Yu (2021) measure daily forced liquidations in bitcoin perpetual futures at 3.51 percent of outstanding long open interest and 1.89 percent of short, on positions averaging roughly sixty times leverage. Brunnermeier and Pedersen (2009) describe the

general form of this feedback, in which falling prices tighten funding constraints which force further selling. The cascade is therefore not a temporary imbalance awaiting an intermediary but a mechanical process that continues until the leverage is gone, and there is no particular reason for it to stop at the close of any given session, because these markets do not close. That account is a candidate explanation and not a demonstrated one, for the reason Section 10 gives: this design observes prices, not sellers. A second and more prosaic difference is that the crypto events are frequent and shallow relative to their own volatility, whereas the surviving equity result lives at a depth that occurs roughly once every seven and a half years. The equity finding is a statement about a rare extreme, and the crypto sample contains almost nothing comparably rare relative to its own distribution.

This asymmetry has a practical implication that generalizes past these two markets. If reversal after forced selling is payment for supplying liquidity, then it should be sought where liquidity provision is genuinely constrained and genuinely compensated, not merely where prices move a great deal. Volatility and opportunity are not the same variable, and a market that produces large declines continuously may produce no reversal at all.

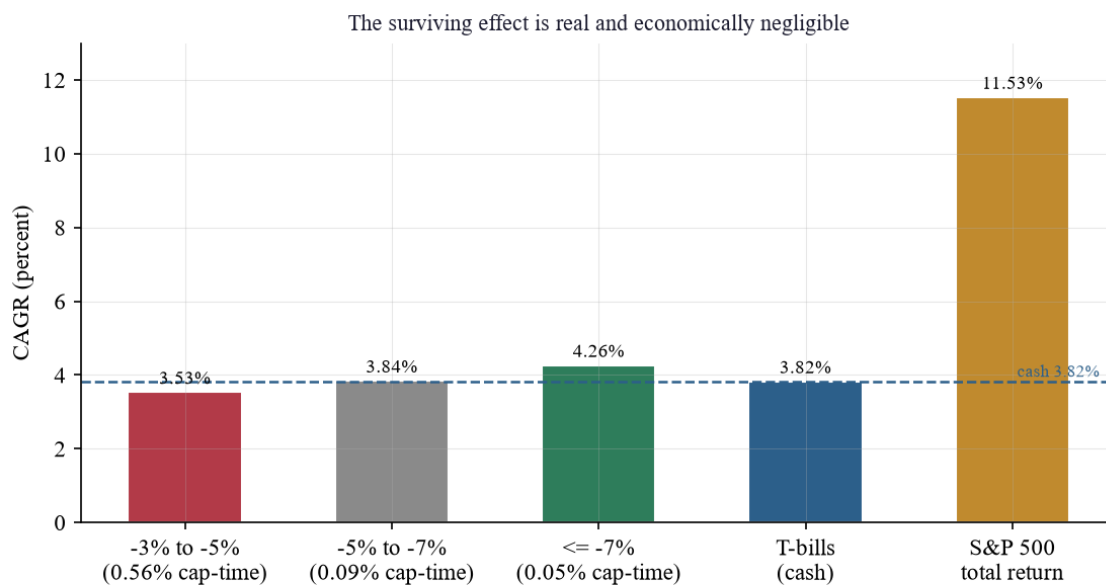
9. The Account: Real and Economically Negligible

A statistical result and a strategy are different objects, and the gap between them is where most published event studies stop. The account simulation walks all 19,276 sessions, deploys on flush dates and holds Treasury bills otherwise. Table 3 gives the cards. The shallowest band compounds to a return of 3.53 percent a year, which is below the 3.82 percent the same capital would have earned sitting in bills, and its Sharpe ratio against that benchmark is negative at minus 0.14. The middle band returns 3.84 percent, which is cash to within two basis points a year. The tail band returns 4.26 percent, beating cash by 0.44 points a year with a maximum drawdown of 2.7 percent.

Band	Active days	Capital time	Day win rate	Mean per active day	CAGR	maxDD	Sharpe
-3% to -5%	322	0.555%	49.7%	-0.050%	+3.53%	13.2%	-0.14
-5% to -7%	55	0.091%	54.5%	+0.056%	+3.84%	5.5%	0.02
At or below -7%	25	0.052%	72.0%	+1.349%	+4.26%	2.7%	0.31
Treasury	n/a	0% (all	n/a	n/a	+3.82%	0.0%	n/a

bills		cash)					
S&P 500 price only, 1950 to 2026	n/a	100%	n/a	n/a	+8.33%	56.8%	0.35
S&P 500 total return, 1988 to 2026	n/a	100%	n/a	n/a	+11.53%	55.3%	0.54

Table 3. Account cards over the fixed calendar of 19,276 sessions from 1950 to 2026, unlevered, twenty basis points round trip, twenty percent per position, idle cash earning the Treasury bill rate. The strategy rows and the Treasury bill row cover the full calendar; the total return row begins in 1988 where that series does, and the price only series is shown on the full calendar for a like for like span. Neither equity row is an alternative at equal risk, since both are held at one hundred percent capital time against the tail band's 0.052 percent. Capital time is the fraction of dollar days actually at risk, which is the honest exposure unit; counting active days instead would overstate it by a factor of about three.



The tail overlay beats cash by 0.44 points a year while deploying 0.05 percent of capital time. It never approaches equities.

Figure 5. Compound annual return by band against cash and equities. The shallowest band underperforms Treasury bills outright. The tail band beats bills by 0.44 points a year while deploying 0.052 percent of its capital time, and nothing in the family comes close to simply owning the index.

The capital time column is the one that settles the practical question. The tail strategy is at risk on 0.052 percent of dollar days. Per unit of deployed capital the return is very large, 1.349 percent on an active day against an unconditional daily mean near 0.03 percent, which is the

correct way to describe the edge and is also why the effect is worth documenting. But the account that runs it is a Treasury bill portfolio with a rare overlay, and no amount of skill at the overlay changes the fact that it can be applied twenty five times in seventy six years. This is a capacity and frequency constraint, not a quality constraint, and it cannot be relieved by leverage without reintroducing exactly the tail risk the strategy is trying to harvest.

Two caveats belong with these cards. The reported drawdowns are small partly because the strategy is almost never exposed, so they describe the account rather than the risk of the trade; a single tail event that failed to revert would move them substantially. And the variant that buys the next open, which is the fully implementable one, produces materially the same cards, with the tail band at 4.16 percent and an identical Sharpe ratio of 0.31, so the result does not depend on the unrealistic assumption of transacting at the closing print.

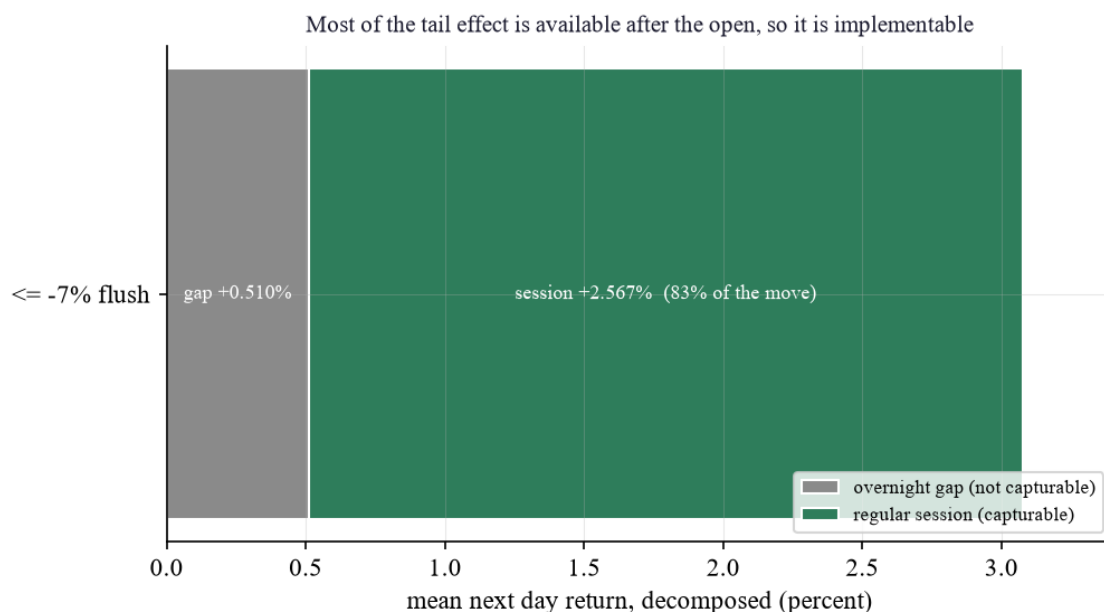


Figure 6. The tail move decomposed into the overnight gap and the regular session. Eighty three percent of the average next day return accrues after the opening print, so a participant who waits for the open and buys there captures most of the effect. This is what makes the result a finding rather than a curiosity.

10. Discussion and Limitations

The strongest and most transportable result in this paper is the counting correction, because it applies to any event study in which events cluster in calendar time. A pooled t of 6.76 that becomes 1.68 is not an unusual case; it is the expected consequence of sampling twenty eight

instruments that fall together, and it is the over rejection that Kolari and Pynnonen (2010) quantify. The same logic covers the nesting of thresholds and the choice of block length, and the general lesson is that in event studies the effective sample size is almost always smaller than the row count, sometimes by an order of magnitude. The companion papers in this series make the same argument in different settings (Dashyan, 2026a, 2026b, 2026c, 2026d).

The surviving positive result should be held with correspondingly loose hands. It rests on 25 dates clustering into 10 episodes across seventy six years, and while it clears a familywise correction and survives the removal of the two most obvious years, no procedure can manufacture independent evidence that history did not supply. Ten episodes is a small number for a claim about a rare regime, and the confidence interval, from 1.34 to 4.84 percent, is correspondingly wide. A reader who concluded only that something happens after seven percent index declines and that its size is poorly determined would be reading the evidence correctly.

Several limitations belong in the open. The single name panel is survivorship biased, and although the bias runs against the null we report, a properly constructed panel including delisted securities would be a better test. The crypto comparison is cited from an earlier stage of this program rather than recomputed here, and its pre 2026 events are identified from price and volume because a native liquidation feed did not exist; that is a meaningful weakness in the identification of the event, though not one that plausibly turns a uniformly negative result positive. Both asset classes are tested with a single crude event definition and a one session horizon, which is a deliberate choice to avoid specification search but leaves open whether a more carefully constructed signal would do better. And the comparison across asset classes is not like for like in an important respect: the equity test spans seventy six years and the crypto test six, so the crypto null is a strong statement about a short period rather than a weak statement about a long one.

Finally, a word on the relationship between this paper's two halves. It would be neater if the equity result and the crypto result had a single explanation, and Section 8 offers the most plausible candidate. But the honest position is that this paper establishes the asymmetry more firmly than it explains it. Distinguishing the automatic liquidation account from the simpler observation that crypto markets never close, or from the possibility that the crypto sample

contains no event as rare as a seven percent index day, would require a research design this paper does not have.

11. Conclusion

Buying after a large decline is a good idea that mostly does not work, and the gap between the two is largely a matter of how the evidence is counted. Once events that share a calendar date are counted once, once depth thresholds are made disjoint rather than nested, once the several thresholds tested are charged a familywise correction, and once the bootstrap is calibrated rather than assumed, the effect disappears from every setting examined here except one. What remains in equities is a large next session return following index declines at or below seven percent, concentrated in a single session rather than spread across a recovery window, available to a participant who waits for the open, and resting on twenty five dates in seventy six years. What remains in crypto perpetual futures is nothing at all: an edge that is negative in every regime and every year, that a machine learning model cannot find, that does not improve when inverted, and that does not respond to changes in geometry. And what remains for an investor is a Treasury bill account with an overlay that fires once every seven and a half years and adds less than half a point a year, which is the difference between establishing that an effect exists and establishing that it is worth trading.

Author Contributions and Use of AI Tools

The research questions, experimental design, hypotheses, methodological choices, and interpretation of results in this paper are the author's own. All software implementation, including the event study harness, the bootstrap and familywise corrections, the placebo and decomposition tests, the account simulator, the adversarial verification harness, and the figures, was written in Python with the assistance of Claude, a large language model by Anthropic, acting as a coding assistant under the author's direction. The verification process that identified and corrected the block bootstrap miscalibration, the nested threshold problem, and the missing multiplicity control described in Section 4 was itself run as a multi agent process using the same tool. Claude is a software tool and is not an author of this work. The author is solely responsible for the content, the claims, and any errors.

Data and Code Availability

The equity analysis is fully reproducible. Prices are daily open and close series retrieved from a public vendor chart API for four indices and twenty eight single names, together with the thirteen week Treasury bill series; a single deterministic script with a fixed random seed produces every equity number, table and figure in this paper, including the event study, the familywise correction, the placebo horizon scan, the gap and session decomposition, and the account simulation. A second deterministic script produces the calibration evidence reported in Section 4, namely the Monte Carlo size of the bootstrap under both block settings, the autocorrelation and event spacing diagnostics, and the standardized threshold comparison of Section 6. An earlier version of the same script, which used the nested thresholds and the miscalibrated block bootstrap described in Section 4, reproduced every t statistic reported in the companion methodology paper (Dashyan, 2026b) to within 0.03, the one decimal place at which that paper reported them, which is how the estimator was validated before the design was corrected. The crypto results in Section 7 are cited from a dated internal record of the research program rather than recomputed for this paper: of the scripts that produced them only the bracket geometry retest survives in the repository, the others having been written in temporary session workspaces, and readers should treat those figures as reported results with a documented provenance rather than as independently reproducible artifacts. The scripts, the numeric ledger underlying every figure, and the verification transcripts are available from the author on reasonable request, subject to the terms of the third party data sources.

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